

Lecture : DP in practice -II: Collecting telemetry data privately

[Ding, Kulkarni, Yekhanin (2017)]

In this lecture, we shall discuss a communication-efficient LDP algorithm for estimating the mean of "telemetry data" $\equiv$ data mined from users of mobile phones, for example. The mechanism we shall discuss is a 1-bit (!) mechanism for computing an estimate of the mean, and has been deployed by Microsoft.

As before, let $(v_i : i \in [N])$, where $v_i \in [0, m]$ denote the items with users $[N]$. We are interested in estimating

$$\mu \triangleq \frac{1}{N} \sum_{i=1}^N v_i.$$

via an ϵ -LDP mechanism.

Algorithm 1Bit (Users)

→ Given $v_i \in [0, m]$, each user releases

$$b_i = \begin{cases} 1, & \text{w.p. } \frac{1}{e^{\epsilon} + 1} + \frac{v_i}{m} \cdot \frac{e^{\epsilon} - 1}{e^{\epsilon} + 1}, \\ 0, & \text{o.w.} \end{cases} \quad \text{--- (1)}$$

to the central server.

Algorithm (Sewer)

→ The sewer computes the mean estimate

$$\hat{\mu} \triangleq \frac{m}{N} \cdot \sum_{i=1}^N \frac{b_i (e^{\epsilon+1}) - 1}{e^{\epsilon} - 1}.$$

Claim 1: Algorithm 1Bit is ϵ -LDP.

Proof: Observe that $P[b_i=0]$ varies from $\frac{1}{e^{\epsilon+1}}$ to $\frac{e^{\epsilon}}{e^{\epsilon+1}}$, and so does

$P[b_i=1]$. Thus, we have that $\frac{P[b_i=0]}{P[b_i=1]} \leq e^{\epsilon}$ and $\frac{P[b_i=1]}{P[b_i=0]} \leq e^{\epsilon}$. $\square$

Claim 2: $\hat{\mu}$ is an unbiased estimator of μ .

Proof: We have that

$$E[\hat{\mu}] = \frac{m}{N} \cdot \frac{e^{\epsilon+1}}{e^{\epsilon}-1} \left(\sum_{i=1}^N E[b_i] - \frac{N}{e^{\epsilon+1}} \right)$$

$$= \frac{m}{N} \cdot \frac{e^{\epsilon+1}}{e^{\epsilon}-1} \left(\sum_{i=1}^N \frac{v_i}{m} \cdot \frac{e^{\epsilon}-1}{e^{\epsilon+1}} \right)$$

$$= \frac{1}{N} \sum_{i=1}^N v_i = \mu. \quad \square$$

Claim 3: For any $\beta \in (0,1)$, we have

$$P_h \left[|\hat{\mu} - \mu| \geq \beta m \right] \leq 2 \cdot \exp \left(-2\beta^2 N \cdot \left(\frac{e^\epsilon - 1}{e^\epsilon + 1} \right)^2 \right).$$

Proof: Via an application of Hoeffding's inequality to RVs bounded in $[0,1]$, we have for any $t \geq 0$,

$$P_h \left[\left| \sum_{i=1}^N b_i - \frac{N}{e^\epsilon + 1} - \frac{N\mu}{m} \cdot \frac{e^\epsilon - 1}{e^\epsilon + 1} \right| \geq t \right] \leq 2 \exp \left(-\frac{2t^2}{N} \right).$$

Rearranging, we see that

$$P_h \left[|\hat{\mu} - \mu| \geq \frac{tm}{N} \cdot \frac{e^\epsilon + 1}{e^\epsilon - 1} \right] \leq 2 \cdot \exp \left(-2t^2/N \right).$$

Setting $t = \beta N \frac{e^\epsilon - 1}{e^\epsilon + 1}$, we obtain the statement of the claim. $\square$

As an immediate corollary of Claim 3 above, we obtain the following result.

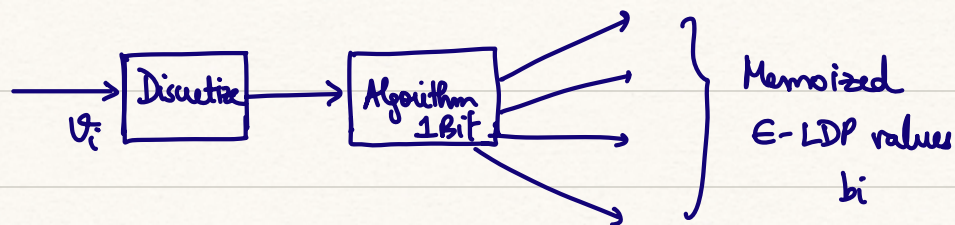
Theorem 1: The ϵ -LDP mechanism Algorithm 1Bit satisfies, w.p. $\geq 1-\delta$, for $\delta \in (0,1)$,

$$|\hat{\mu} - \mu| \leq \frac{m}{\sqrt{2N}} \cdot \frac{e^\epsilon + 1}{e^\epsilon - 1} \cdot \sqrt{\log \left(\frac{2}{\delta} \right)}.$$

While Algorithm 1Bit is useful for a single collection of bit estimates for mean computation, in practice, users are queried several times over for data mining, which, by the Basic Composition Theorem, results in a privacy loss that grows linearly in the # of queries.

A simple trick to avoid this growing privacy loss is memoization: release the same value if the same query is repeated several times. Clearly, since no fresh randomness is used in subsequent responses, the overall privacy loss remains ϵ .

However, for memoization to be applicable in practice, we must have a discrete input domain. Hence, we must pre-process inputs to discretize them, then apply Algorithm 1Bit with memoization. The pipeline hence is:



We now discuss a discretization procedure that, perhaps surprisingly, provides the same accuracy guarantees as Theorem 1 (applied to non-discretized values).

Algorithm Round (User)

→ Pick a large integer $s|m$.

→ Each user $i \in [N]$ independently picks $\alpha_i \in \{0, 1, \dots, s-1\}$. Let $x_i \in [L, R)$, where $L, R \in \{ls\}_{l=0}^{l=m/s}$. The user i rounds v_i to L if $v_i + \alpha_i < R$, and to R , otherwise.

Let $y_i = \text{Round}(v_i)$ and let $\mu' = \frac{1}{N} \sum_{i=1}^N y_i$.

Claim 4: We have $E[\mu'] = \mu$.

Proof: One can write $y_i = v_i + Z_i$, where $Z_i = \begin{cases} R - v_i, & \text{w.p. } \frac{v_i - L}{R - L}, \\ v_i - L, & \text{o.w.} \end{cases}$.

Hence, since $E[Z_i] = 0$, we have $E[y_i] = v_i$, implying that $E[\mu'] = \mu$. $\square$

Theorem 2 (informal): Algorithm 1Bit, when applied to the discretized values $\{y_i\}$ has the same accuracy guarantees as in Theorem 1, independent of s .

Proof: It suffices to show that, for all s , each output bit b_i returned by Algorithm 1Bit has the same distribution as in (i).

To see why, let $b \triangleq R - v_i$ and $a \triangleq v_i - L$. Then, we have post discretization

$$P_i[b_i = 1]$$

$$= \frac{b}{a+b} \left(\frac{1}{e^{\epsilon+1}} + \frac{L}{m} \cdot \frac{e^{\epsilon}-1}{e^{\epsilon+1}} \right) + \frac{a}{a+b} \left(\frac{1}{e^{\epsilon+1}} + \frac{R}{m} \cdot \frac{e^{\epsilon}-1}{e^{\epsilon+1}} \right)$$

$$= \frac{1}{e^{\epsilon+1}} + \frac{v_i}{m} \cdot \frac{e^{\epsilon}-1}{e^{\epsilon+1}}, \text{ after some algebra. } \square$$